# Venkata Lakshmi Parimala Pasupuleti

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## SUMMARY

Backend software engineer with 6+ years building data-intensive production systems, AI agents, and ETL pipelines. At Intuit, building agentic AI workflows using the Anthropic Messages API and Agent SDK, integrated into production financial data services. Built Leasel -- an 8-agent LangGraph orchestration system with 93.6% precision across 1,240 test cases, applying the same agent loop pattern used in automated due diligence. Strong Python, SQL, FastAPI, AWS, Kubernetes, Terraform. Background includes ETL pipelines processing 1.6M+ records, 200+ SQL query optimization across multi-tenant PostgreSQL/Oracle, and async processing at scale with idempotency guarantees. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## CORE COMPETENCIES

Python / FastAPI / SQL | TypeScript / JavaScript | AI Agents / Agent SDK / LangGraph | Messages API (Anthropic Claude)

ETL Pipelines / Data-Intensive Systems | PostgreSQL / Advanced SQL | Async Processing (RabbitMQ)

AWS (Lambda, S3, RDS, SQS, API Gateway) | Kubernetes | Terraform (IaC) | Docker / Microservices

REST APIs / Backend Infrastructure | ML Pipelines (scikit-learn, XGBoost) | CI/CD (GitHub Actions)

Observability (Prometheus, OpenTelemetry) | Caching / Redis

## EXPERIENCE

### Intuit

Jun 2026 -- Present

**Software Engineer** -- Virginia, USA

- Build production agentic AI workflows using the **Anthropic Messages API and Agent SDK** -- tool use, structured output, guardrails, and input validation integrated into financial data services.
- Build full-stack features across Python/FastAPI backend and **TypeScript/React frontend**; design and query SQL schemas supporting financial reporting and downstream analytics.
- Build **ML pipelines** (scikit-learn, XGBoost) from ETL and feature engineering through prediction APIs; deploy on AWS Lambda/S3/RDS via Docker and GitHub Actions CI/CD.
- Build golden-dataset regression harnesses to catch agentic AI quality regressions before release; instrument services with Prometheus and OpenTelemetry for production observability.

### Entrata -- Residential & Affordable Housing SaaS

Oct 2020 -- Jul 2024

**Senior Software Engineer** -- Multi-tenant platform, 100K+ monthly active users

- Owned **data-intensive ETL pipelines**: annual HUD income-limit API ingestion transforming federal source data into ~180,000 domain records, with validation, reconciliation, and rollback at each release cycle. Zero data integrity failures over 6 years.
- Built the **eligibility decisioning engine** automating applicant qualification -- AMI threshold calculations, income verification rules, HUD/LIHTC compliance checks, and Approve/Escalate routing across 100K+ units.
- Built **RabbitMQ-based async processing** for rent and eligibility updates across thousands of tenant records -- idempotency guarantees, retry logic, and dead-letter handling in a multi-tenant distributed system.
- Optimized 200+ SQL queries via execution plan analysis -- 30% performance improvement across 1M+ records; wrote complex joins, conditional logic, and stored procedures across multi-schema PostgreSQL/Oracle environments.
- Designed Docker microservices architecture (50% better scalability, 30% faster deployments); led AWS S3 migration with PHI encryption (Libsodium) covering 100% of resident data.
- Built Python/Bash automation reducing manual operational work by 80%; reviewed 300+ PRs; mentored junior engineers on distributed systems and data pipeline patterns.

### Entrata India

Sep 2018 -- Oct 2020

**Junior Software Engineer** -- Pune, India

- Built property management backend modules (Python, Java, PostgreSQL) with data import pipelines processing 1.6M HUD/Tax Credit records; implemented role-based access control and multi-tenant data isolation.

## PROJECTS

### Leasel -- 8-Agent LangGraph Orchestration System

Python · LangGraph · Agent SDK · FastAPI · PostgreSQL

8-agent LangGraph workflow automating housing compliance due diligence: triage, data retrieval, rule evaluation, confidence scoring, and human escalation at defined thresholds. Each agent has a clear goal, tool set, verification step, and stop condition -- the same agentic loop pattern used in automated permitting and siting workflows. LLM-as-judge CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

### AgentProbe -- AI Agent Evaluation & Inference Infrastructure

Python · FastAPI · Provider Routing · CI/CD · Observability

15-endpoint FastAPI service with provider routing across five LLM/ML inference backends. Scoring engine evaluates agent output quality on five metrics; CI regression against golden datasets. Same pattern as production AI agent evaluation infrastructure.

github.com/venkatalakshmiparimala/AgentProbe

### AI on FHIR -- Clinical Decisioning API

Python · Flask · REST APIs · RBAC · Docker · HIPAA

RESTful API translating clinical queries into FHIR data calls using NLP agents. Fine-grained RBAC, HIPAA-style data isolation, and containerized deployment -- regulated-data agentic pattern.

github.com/venkatalakshmiparimala/AI_on_FHIR

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University, Virginia — May 2026 | GPA 3.87

**B.Tech Computer Science & Engineering** — JNTU Anantapur — 2017